Sandeep Kumar

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Education

Indian Institute of Technology Roorkee — Uttarakhand, India
Integrated Dual Degree (Masters+Bachelors) in Applied Mathematics — *2020 – 2025*

Experience

WorldQuant Brain — June 2023 – June 2025
Research Consultant | Financial Analysis, Mathematical Modeling — *Remote*

- Formulated and backtested **50+ statistical arbitrage** strategies using time-series analysis, cross-sectional factor modeling, and mathematical transformations (**z-scores**, **rank correlation**) to extract signals from market data.
- Evaluated and optimized portfolio risk by **neutralizing** factor exposures (sector, style, market) to minimize drawdowns and turnover, while running stress testing and out-of-sample analysis to eliminate overfitting.
- Engineered automated Python workflows on WorldQuant ACE platform for high-throughput **alpha** mining, systematic parameter tuning, and filtering signals via **Sharpe**, **Information Ratio**, and **decay metrics**.

Standard Chartered Bank — July 2025 – Present
Development Analyst | Python, REST APIs — *Bengaluru*

- Developed and maintained a full-stack transaction automation platform (PT HARNESS) using Python, PySpark, React.js, Flask, and REST APIs to streamline enterprise **payment processing** workflows.
- Designed Oracle database-integrated data processing pipelines using Python, PySpark, SQL, and oracledb for transaction tracking, reconciliation, and SLA-based monitoring across OLTP/OLAP systems.
- Streamlined report generation and execution lifecycle management across **20+** enterprise modules, reducing manual monitoring overhead and improving end-to-end testing visibility.

Projects

Connect 4 AI Agent — Aug 2024 – Nov 2024

- Modeled the 6×7 grid as a finite **Markov Decision Process** (MDP), leveraging state space S, action space A, and reward R to evaluate optimal move-state transitions.
- Implemented **Deep Q-Learning (DQN)** for non-linear state-action value approximation achieving a **99.38%** win rate over $\mathbf{4.5 \times 10^{12}}$ game states.
- Minimized **Temporal Difference** (TD) error δ_i via **Huber Loss**, updating network parameters against **Bellman Target**.
- Formulated a linear output layer predicting **7 action Q-values**, using an **exploration-exploitation** policy to drive convergence.

Physics-Informed Neural Networks Solution for Lid-Driven Cavity Flow — Jan 2025 – April 2025

- Solved 2D incompressible **Navier–Stokes equations** for lid-driven cavity flow using a custom Physics-Informed Neural Network (PINN) architecture without relying on external simulation datasets.
- Embedded continuity and non-linear momentum equations into the loss function via **automatic differentiation**, training over **10,000+ collocation points** to minimize PDE residual loss to $< \mathbf{10^{-4}}$.
- Evaluated flow behavior across Reynolds numbers $\mathbf{Re = 100}$–$\mathbf{1000}$, achieving $< 1.5\%$ **relative** L_2 **error** against the **Ghia et al. CFD benchmark** for centerline velocity profiles and primary vortex locations.

Advection–Diffusion–Reaction Equations Using Physics-Informed Neural Networks — Jan 2023 – April 2023

- Engineered a mesh-free **PINN framework** to solve 1D/2D Advection–Diffusion–Reaction (ADR) equations across varying transport regimes, driving PDE loss residual to $< \mathbf{10^{-4}}$.
- Implemented novel collocation sampling methods **First Derivative Constraint (FDC)** and **Orthogonal Grids** reducing relative L_2 error by $\sim \mathbf{45}\%$ compared to standard uniform sampling.
- Utilized **automatic differentiation** to evaluate gradient pathology in advection-dominated flows, achieving $> \mathbf{99}\%$ solution accuracy against analytical reference solutions.

Skills

Languages: C/C++, Python, Java
Relevant Coursework: Probability & Statistics, Linear Algebra, Differential calculus, Stochastic process, Field Theory.
ML/AI: Deep Learning, Generative AI, LLMs, TensorFlow, CUDA, Reinforcement Learning, Applied Mathematics.

Achievements

Codeforces (rating - **1802**, handle: Sandeep5445), Codechef (rating - **2060** , handle: sandeep_1132).
Secured **141st** (2nd in India) position in ICPC 2023 Online Spring Challenge powered by Huawei.
National Finalist – International Quant Championship 2023, WorldQuant Brain.
Secured **Ranks 186 and 1030** in Meta Hackercup Round 3 (2025) and Round 2 (2023), respectively.
Secured **Ranks 17 and 37** in CodeChef Starters 241 and Starters 209, respectively.
Secured **Ranks 64 and 110** in Codeforces Round 867 and Round 1058, respectively.